ANIMAL KINGDOM

Level of Organisation :

Cellular level - Porifera - In sponges the cells are arranged as loose cell aggregates

Tissue level - Cnidaria, Ctenophora Organ level - Platyhelminthes
Organ System level - Aschelminthes - Chordata

* Incomplete digestive system - has single opening to the outside of the body that serves as both mouth and anus.

* Complete digestive system - has two separate opening as mouth and anus.
Aschelminthes - Chordata

Circulatory System

OPEN ← cells and tissues are directly bathed in which the blood that is pumped by heart
Arthropoda, Mollusca, Echinodermata, Hemichordata

CLOSED → Blood is circulated through a series of blood vessels.
Annelida
vertebrates

Symmetry

1) Asymmetrical - Any plane passes through centre and does not divide animals into two equal halves (Porifera)

2) Radial symmetry - Any plane passes through centre and divide animals into two equal halves (Cnidaria, Ctenophora, adult Echinodermata)

3) Bilateral Symmetry - Only one plane divides into right & left identical halves (Platyhelminthes to Chordata)

Symmetry

Asymmetrical
Sponges

Radial
Cnidaria, Ctenophora, Echinodermata

Bilateral
Platyhelminthes to Chordata

Germinal Layers

Diploblastic
- Two layers are fnt Ectoderm and Endoderm
- Coelenterata, Ctenophora

Triploblastic
- Three layers are fnt
- Ectoderm, Endoderm, Mesoderm
Platyhelminthes to Chordata

Coelom

Presence or absence of a cavity b/w the body wall and the gut wall is very important in classification.

Coelom

Acoelomates
If body cavity is absent.
Porifera
Cnidaria
Ctenophora
Platyhelminthes

Pseudocoelomate
Body cavity is not lined by mesoderm, instead, mesoderm is fnt as scattered pouches in b/w the ectoderm & endoderm.
Aschelminthes

Coelomate
Body cavity is lined by mesoderm.
Annelida to Chordata

Segmentation → In some animals, the body is externally and internally divided into segments with a serial repetition of at least some organs.
eg. Earthworm (Annelida), Arthropoda, Chordata

Notochord → Rod like structure formed during embryonic development on the dorsal side

It is mesodermally derived

Animals with notochord → Chordates

Animal do not have notochord → Non-chordates
eg. Porifera to Echinodermata

Phylum - Porifera

- Also called sponges.
- Generally marine and asymmetrical.
- Cellular level of organisation.
- Diploblastic animals
- Food gathering, Respiratory exchange and removal of waste occur through water canal system.
- Digestion intracellular.
- Ostia (minute pore on body), spongocoel (body cavity) and osculum help in water transport
- They are lined by choanocytes (collar cells)
- Body wall has skeleton of spicules or spongin fibres.
- sexes are not separate (hermaphroditic)
- Fertilisation is internal

example - Sycon, spongilla (Fresh water sponge) and Euspongia (Bath sponge)

Coelenterata (Cnidaria)

- Cnidoblast / Cnidocytes → used for anchorage,

(contain stinging capsule) defense, and capture of prey.

- They have gastro-vascular cavity with a single opening.
- Mouth on Hypostome.
- 2 Basic Body forms:
  Polyp → sessile & cylindrical (ex. Hydra & Adamsia)
  Medusa → free swimming & Umbrella shaped (Aurelia or Jelly fish)
- Metagenesis → i.e. Polyp produce Medusa by asexually and Medusa produce polyp sexually (e.g. - Obelia)

Ctenophora (sea walnut / comb jellies)

- Exclusively Marine
- Body bears 8 external rows of ciliated comb plates, help in locomotion.
- Bioluminescence is well marked.
- Sexes are not separate
- Reproduction - Only Sexual
- Radial symmetrical, diploblastic organisms

Platyhelminthes (Flatworms)

- Dorso-ventrally flattened body called flatworms
- Mostly endoparasites found in animals including, human beings
- Specialised cells called flame cells helps in osmoregulation & Excretion.

- Sexes are not separate.
  They have high regeneration capacity like Planaria

Aschelminthes (Roundworms)

- Circular in cross section
- They may be free living, aquatic & terrestrial or parasitic in plant & animals.
- well developed muscular pharynx
- Sexes are separate.
- Often female are longer than male.
- Complete Alimentary canal.

Annelida

- They may be aquatic (marine and freshwater) or terrestrial, free living and sometimes parasitic.
- Their body surface is marked out into segments and metameres.
- They posses longitudinal and circular muscle which help in locomotion.
- Nereis posses lateral appendages, called parapodia which help in swimming.
- Circulatory System - Closed type
- Nephridia - Help in osmoregulation & excretion
- Neural system consists of paired ganglia connected by lateral nerves to a double ventral nerve cord.

Arthropoda (Largest Phylum)

- Over two-thirds of all named species are arthropods
- Body consists of Head, Thorax and abdomen
- They have jointed appendages (arthro-joint,

poda - appendages)

- Sensory organs like antennal, eye, statocysts or balancing organs are present.
- Body of arthropods is covered by chitinous exoskeleton
- Respiratory organs - Gills, Book gills, Book lungs or Tracheal System.
- Excretion takes place through malpighian tubules

Mollusca (2nd largest Phylum)

- They are terrestrial or aquatic.
- Body is covered by a calcareous shell and is unsegmented with a head, muscular foot and visceral hump
- Soft layer of skin forms Mantle over the visceral hump
- Space b/w Hump & Mantle is called Mantle cavity in which feather like gills are ent. They have respiratory and excitatory functions.
- Anterior region has sensory tentacles.
- Mouth contains file like rasping organ for feeding called Radula.

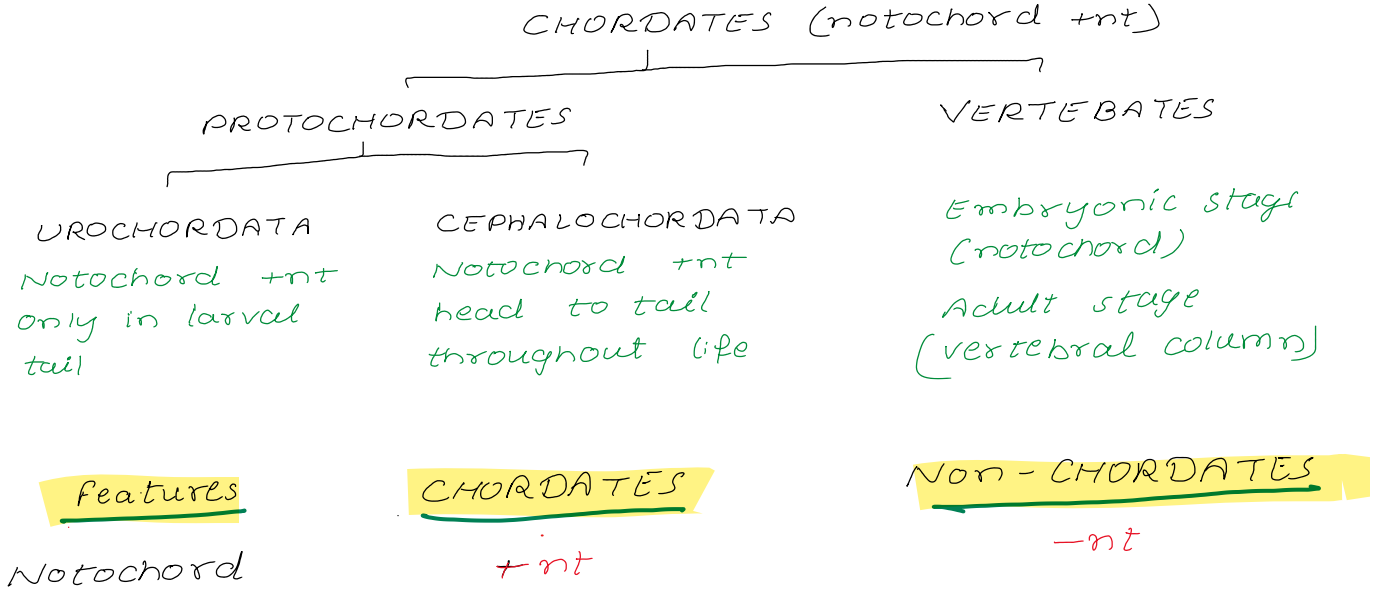
Echinodermata (spiny bodied animals)

- All are marine.
- Mouth - lower side (ventral), Anus - upperside (Dorsal)
- They have water vascular system which help in locomotion, food capturing, Respiration
- Reproduction is sexual; sexes are separate
- Excretory system absent.
- Organ System level of organization.

Hemichordata (worm-like Marine animals)

- Earlier considered as subphylum under 'Chordata' Now separate phylum under non chordates.

- Body is composed of Proboscis, Collar and trunk.
- Rudimentary structure in collar region called Stomochord (similar to notochord)
- Respiration takes place through gills.
- Excretory organ is proboscis gland.
- Sexes are separate. Fertilisation is External.



Gill-slits	+nt	-nt
Post-anal tail	+nt	-nt
Heart	Ventral	Dorsal
CNS	Dorsal, Hollow, Single	Ventral, Solid, Double

- Vertebrata
  - Agnatha (lack jaw)
    - Class - Cyclostomata
  - Gnathostomata (bear jaw)
    - Pisces (bear fins)
      1) Chondrichthyes
      2) Osteichthyes
    - Tetrapoda (bear limbs)
      1) Amphibia
      2) Reptilia
      3) Aves
      4) Mammals

Class - Cyclostomata

- All are Ectoparasites on some fish.
- Body bears 6-15 pair of gill slits.

- Sucking and circular mouth (Jaw absent)
- Scales and paired fins absent.
- Cyclostomes are marine but migrate for spawning to fresh water
- Their larval to marine water.

Features	Chondrichthyes	Osteichthyes
Endoskeleton	Cartilaginous	Bony
operculum	Absent	Present
Air Bladder	Absent	Present
Oviparous/Vivi-parous	Viviparous	Oviparous
Mouth	Ventral	Terminal
Scales	Placoid	Cycloid/Ctenoid

Chondrichthyes ← Electric Organ (ex - Torpedo)
Poisonous sting (ex - Trygon)

- Chondrichthyes are poikilothermous (cold blooded)
- In males pelvic fins bear claspers.
- They have internal fertilisation.

Class - Amphibia Amphi (dual + Bios (life) → Aquatic

+ terrestrial)

- Body is divisible into Head & trunk.
- Skin (moist), Eyes have eyelids, Tympanum represents Ear.
- Alimentary canal, Urinary and Reproductive tracts open into a common chamber called cloaca.
- The Heart is 3-chambered (2 auricle one ventricle)
- Oviparous animals, Development - Indirect.

Class - Reptilia

- Body is covered by dry & cornified skin, epidermal scales/scutes.
- External ear opening Absent, Tympanum represent Ear.
- Heart is 3 chambered, (exception Crocodile - 4 chambered Heart)
- Snakes and Lizards shed their scales as skin cast.

Class - Aves (Birds)

- Feathers and Beak ↑nt, most of them fly except Ostrich
- Forelimbs are modified into wings. Hind limbs have scales and are modified for walking, swimming or clasping tree branch.
- Skin is dry without glands except oil glands at the base of tail.
- Endoskeleton is fully ossified (bony)
- Long bones are hollow with Air cavities (Pneumatic)
- Digestive tract of birds has additional chambers the crop & Gizzard; Heart - 4 chambered, warm blooded
- Respiration is by Lungs, Air sacs connected to lungs supplement respiration. Sexes are separate.
- Fertilisation - Internal
- Oviparous and development is direct.

Class - Mammalia (↑nce of mammary glands

- Skin with Hairs
- External Ear / Pinnae ↑nt
- Teeth ↑nt → Mostly viviparous

- Heart is four chambered

Examples :

1) Porifera - Sycon (scypha), Spongilla (Fress water sponge), Euspongia (Bath sponge)

2) Coelenterata - Physalia (Portuguese Man-of-war), Adamsia (sea anemone), Meandrina (Brain Coral) Gorgonia (sea-Fan) Pennatula (sea pen),

3) Ctenophora - Pleurobrachia and Ctenoplana

4) Platyhelminthes - Taenia (Tapeworm), Fasciola (Liver Fluke)

5) Aschelminthes - Ascaris (Roundworm), Wuchereria (Filarial worm), Ancylostoma (Hook worm)

6) Annelida - Nereis, Pheretima (Earthworm), Hirudinaria (blood sucking leech)

7) Arthropoda - Economically important insects – Apis (Honey bee) Bombyx (Silkworm), Laccifer (Lac insect)

Vectors - Anopheles, Culex and Aedes (Mosquitoes)

Gregarious Pest - Locusta (Locust)

Living fossil - Limulus (King crab)

8) Mollusca - Pila (Apple snail), Pinctada (Pearl Oyster), Sepia (Cuttle fish), Dentalium (Tusk shell) and Octopus (Devil fish), Chaetopleura (Chiton)

9) Echinodermata - Asterias (star fish), Echinus (sea urchin), Cucumaria (sea cucumber), Ophiura (Brittle star), Antedon (Sea lily)

10) Hemichordata - Balanoglossus and Saccoglossus

11) Urochordata - Ascidia, Salpa, Doliolum.

12) Cephalochordata - Branchiostoma (Amphioxus or Lancelet)

13) Cyclostomata - Petromyzon (Lamprey) & Myxine (Hag fish)

14) Chondrichthyes - Scoliodon (Dog fish), Pristis (saw fish), Carcharodon (Great white shark), Trygon (Stingray)

15) Osteichthyes - Marine - Exocoetus (Flying fish), Hippocampus (sea Horse), Freshwater - Labeo (Rohu), Catla (Katla), Clarias (Magur): Aquarium - Betta (Fighting fish) Pterophyllum (Angel fish)

16) Amphibia - Bufo (Toad), Rana (Frog), Hyla (Tree frog, Salamandra (Salamander), Ichthyophis (Limbless Amphibia)

17) Reptilia – Chelone (Turtle), Testudo (Tortoise), Chameleon (Tree lizard) Calotes (Garden Lizard), Crocodilus (Crocodile), Hemidactylus (Wall Lizard), Poisonous snakes - Naja (Cobra) Bangarus (Krait), Vipera (Viper)

18) Aves – Corvus (Crow), Columba (Pigeon), Psittacula (Parrot), Pavo (Peacock), Apteroclytes (Penguin), Struthio (ostrich), Neophron (vulture)

19) Mammalia – Oviparous – Ornithorhynchus (Platypus)
Viviparous - Macropus (Kangaroo), Macaca (Monkey), Rattus (Rat), Canis (Dog), Equus (Horse), Delphinus (common Dolphin), Panthera Leo Lion). Camelus (camel) Balaenoptera (Blue whale), Panthera tigris (Tiger) Pteropus (Flying fox) Felis (cat), Elephas (Elephant),